

- 4 The *HFE* gene codes for the HFE protein, which has a role in the regulation of iron absorption by the body. Iron is an essential mineral that can be obtained only from the diet.

A mutation of the *HFE* gene known as C282Y causes hereditary haemochromatosis, which is an autosomal recessive disease. The mutant allele codes for a non-functioning protein. People who are homozygous for the mutant allele produce no functioning HFE protein and this results in an excess of iron being absorbed by the body. The accumulation (build-up) of iron in body organs over many years can cause organ damage.

People that are heterozygous for the *HFE* gene do **not** have hereditary haemochromatosis. They do absorb more iron from their diet than people who do **not** have the mutation, but this does **not** usually have any health effects.

- (a) Construct a genetic diagram of a monohybrid cross to show how two parents who do **not** have hereditary haemochromatosis can produce a child with the disease.

Use the following symbols:

H = normal *HFE* allele

h = mutant *HFE* allele.



(b) Some scientists believe that the C282Y mutation may have first occurred in Ireland.

Scientists sequenced DNA obtained from two human fossil skeletons in Ireland. One of the fossils was 5200 years old and the other was 4000 years old.

The scientists concluded that:

- the human living 4000 years ago did have the C282Y mutation
- the human living 5200 years ago did **not** have the C282Y mutation.

Explain how analysis of the results of these DNA sequencing studies could have been carried out to allow the scientists to make these conclusions.

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- (c) At about the same time as the C282Y allele is thought to have first occurred in Ireland, the lifestyle of people in Europe began to change from hunter-gatherers to farmers.

Hunter-gatherers ate mainly meat with some wild plant food. Their diets had high quantities of iron. Early farmers ate mainly plants with some meat. Their diets had lower quantities of iron and these quantities were often inadequate (**not** enough).

Fig. 4.1 is a map of Europe showing the percentage of people in different countries who **now** have one C282Y allele of the *HFE* gene.



Fig. 4.1

Fig. 4.1 shows that the C282Y allele does **not** occur only in Ireland and is now present throughout Europe. The C282Y allele has been maintained in European populations, even though it is a cause of hereditary haemochromatosis.

- (i) Suggest how the C282Y allele of *HFE* has been maintained in European populations.

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- (ii) Suggest explanations for the differences in the percentage of people in different European countries who have one C282Y allele of the *HFE* gene, as shown in Fig. 4.1.

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- (d) Biological databases contain DNA sequence data from a large number of different people.

Table 4.1 shows three of these databases and the percentage of people in each database who have one C282Y allele of the *HFE* gene.

Table 4.1

database	percentage of people who have one C282Y allele of the <i>HFE</i> gene
database A	2.6
database B	9.1
database C	6.3

Suggest **one** reason for the differences between the three databases shown in Table 4.1.

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- (e) In one population consisting of 2501 people, there were 9 people who were homozygous recessive for the *HFE* gene.

Use equation 1 and equation 2 of the Hardy–Weinberg principle to calculate the number of people in the population who are heterozygous for the *HFE* gene.

equation 1: $p + q = 1$

equation 2: $p^2 + 2pq + q^2 = 1$

key to symbols:

p = frequency of the dominant allele

q = frequency of the recessive allele

p^2 = frequency of homozygous dominant genotype

$2pq$ = frequency of heterozygous genotype

q^2 = frequency of homozygous recessive genotype

number of people who are heterozygous for the *HFE* gene = [2]